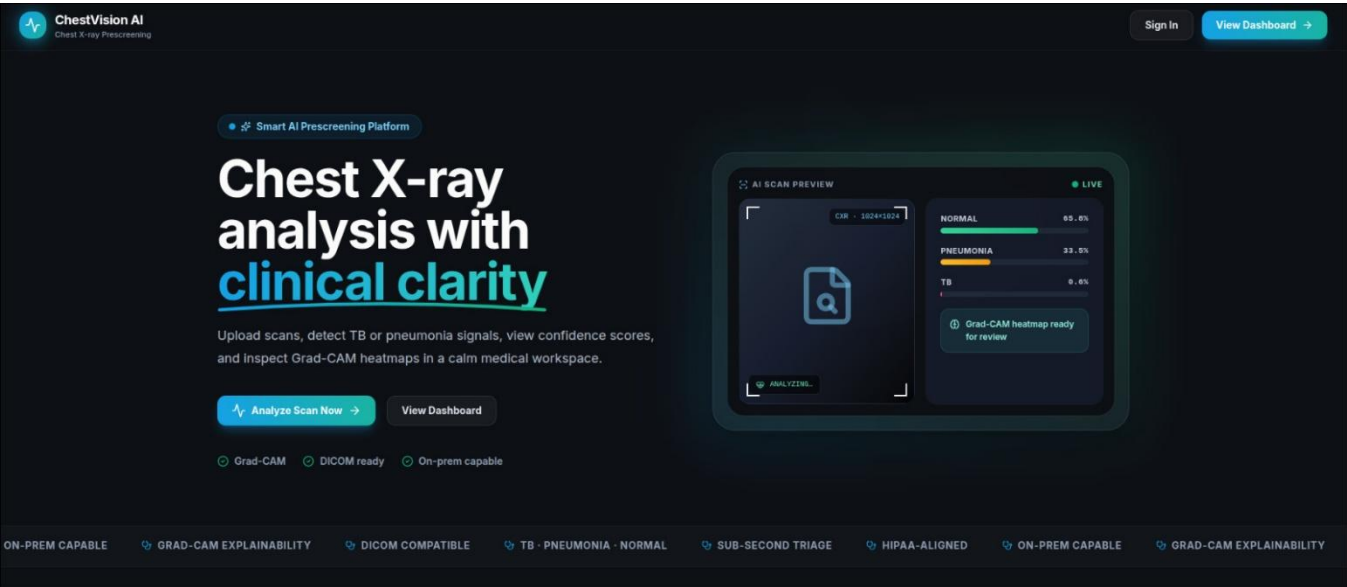


TECHNICAL DOCUMENTATION

ChestVision AI

X-Ray Reader

AI-powered pre-screening of Normal, Pneumonia & Tuberculosis
with Grad-CAM explainability for resource-limited settings



Innovation Theme	Next-Gen Telehealth & AI
Team	Team 42
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Event	DNA Hack For Health 2026 · Chittagong Division · 9 May 2026

1. Problem Statement

1.1 The Diagnostic Gap in Bangladesh

Bangladesh carries one of the world's highest tuberculosis burdens, reporting over 375,000 new TB cases in 2022 — placing it among WHO's 30 high-burden countries that collectively account for 90% of global TB incidence. Pneumonia remains the leading cause of under-5 mortality. Yet both conditions are treatable when detected early. The central failure is not medical — it is diagnostic.

375,000+TB cases Bangladesh (2022)
WHO high-burden country**0.49**Radiologists per 100,000 people
WHO recommends 2.5**2–4 hrs**Travel to nearest radiologist (CHT)
Coastal & hill tracts

1.2 Root Cause

The radiologist shortage cannot be closed by hiring alone. At current medical training rates, Bangladesh would need decades to reach WHO's recommended ratio. The solution must be technological — AI-assisted pre-screening that extends diagnostic reach to district hospitals, upazila health complexes, and community health workers without requiring a specialist on site.

1.3 Target Communities

High-risk populations

- Fishing communities along the Cox's Bazar coastal belt — no specialist access within practical distance
- CHT ethnic minorities in Bandarban, Rangamati, and Khagrachhari — remote terrain, limited infrastructure
- Port workers and EPZ factory workers in Chittagong — high occupational exposure
- Patients at district hospitals served by doctors without radiology support

Facility-level gap

- Bangladesh has ~38,000 upazila and union health facilities
- Most district hospitals have an X-ray machine but no dedicated radiologist
- X-ray images are often read by general physicians without radiology training
- Sputum GeneXpert (gold standard for TB) takes 2 hours and requires lab infrastructure unavailable in most upazilas

ChestVision AI does not replace the radiologist. It gives the general physician a second opinion — flagging cases that need urgent specialist attention — so that no TB case is sent home undetected because the only doctor available was unsure.

2. Proposed Solution

2.1 Overview

ChestVision AI is a web-based AI tool that accepts a chest X-ray image, runs it through a fine-tuned DenseNet121 deep learning model, and returns three outputs: (1) a class prediction — Normal, Pneumonia, or TB; (2) a confidence score and per-class probability breakdown; and (3) a Grad-CAM heatmap overlay showing which lung regions drove the prediction.

2.2 Component Breakdown

Layer	Technology	Details
Frontend	React / HTML	Drag-and-drop X-ray upload, confidence bar display, Grad-CAM 3-panel viewer, clinical advisory notice
Backend API	Python / Flask	inference.py module, validator.py input guard, JSON response with base64-encoded Grad-CAM image
Database	MongoDB / Node.js	Scan history storage, audit log, session management
ML Model	PyTorch + TorchXRayVision	DenseNet121 with chest-specific pretrained weights, Grad-CAM hook on final convolutional layer
Deployment	Kaggle P100 GPU (training)	16GB VRAM Tesla P100; inference runs on CPU or GPU server

2.3 Input Validation Pipeline

Before any image reaches the model, a three-stage validator rejects non-X-ray inputs:

- Aspect ratio check — chest X-rays are roughly square (0.6–1.6 ratio); portrait photos, documents, and screenshots are rejected
- Grayscale dominance test — X-rays are near-perfectly greyscale; colour photographs (R/G/B channel cross-standard-deviation > 15) are rejected
- Histogram distribution test — X-rays have substantial dark pixels (air regions) and lack mid-tone dominance; blank images and colour photos fail this check

Tested against: photograph of a human hand (correctly rejected — colour), blank white image (correctly rejected — histogram), and a valid chest X-ray (correctly passed). The validator runs in under 5ms and adds no perceptible latency.

3. Dataset

3.1 Source Datasets

Dataset	Source	Format	Size	Notes
NIH Chest X-Ray (Pneumonia)	Kaggle / NIH	JPEG	5,216 images	NORMAL + PNEUMONIA classes; US hospital digital radiography
TB Chest Radiography DB	Kaggle	PNG	700 TB + 3,500 Normal	Diverse acquisition settings; includes South Asian patients

3.2 Dataset Engineering

Neither dataset alone provides all three required classes. We engineered a combined dataset through the pipeline:

NIH Chest X-Ray (Pneumonia)	TB Chest Radiography Database	Combined Database
Source: National Institute of Health	Source: Kaggle	Images: Normal + TB + Pneumonia
Images: NORMAL + PNEUMONIA (JPEG)	Images: Normal + Tuberculosis (PNG)	Size: 5083 Normal + 700 TB + 4273 Pneumonia
Original size: 5,216 images	Original size: 700 TB + 3,500 Normal	
Acquisition: US hospital digital radiography	Acquisition: diverse clinical settings. This includes Bangladeshi patients	

Key engineering decisions

- The original pneumonia dataset val/ folder contained only 8 images per class — statistically meaningless. All splits were rebuilt from scratch by pooling the full dataset and re-splitting uniformly
- Split ratio: 75 / 10 / 15 (train / val / test) — weighted toward test to give the minority TB class more test coverage (105 test samples vs 70 at 80/10/10)
- Image mode audit: 57% of images were greyscale (PIL mode 'L'), 41% RGB — the preprocessing pipeline handles both
- Zero corrupt files confirmed across all 10,056 images

3.3 Final Dataset Statistics

Split	NORMAL	PNEUMONIA	TB	Total
Train (75%)	3,812	3,204	525	7,541
Val (10%)	508	427	70	1,005
Test (15%)	763	642	105	1,510
Total	5,083	4,273	700	10,056

3.4 Class Imbalance Handling

TB constitutes only 6.96% of the combined dataset — a 1:7 ratio against NORMAL. Unaddressed, this causes the model to rarely predict TB. Two mechanisms were applied simultaneously:

WeightedRandomSampler

Applied at the DataLoader level. Each sample is assigned a weight inversely proportional to its class frequency. Every training batch draws approximately equal numbers from all three classes, regardless of dataset proportions. TB images are oversampled $\sim 7\times$ per epoch.

Weighted CrossEntropyLoss

Loss function weights: NORMAL $\times 1.0$, PNEUMONIA $\times 1.0$, TB $\times 7.0$ (inverse frequency, normalised). A TB misclassification incurs $7\times$ the gradient penalty of a NORMAL misclassification, forcing the model to prioritise TB accuracy even when the sampler produces a balanced batch.

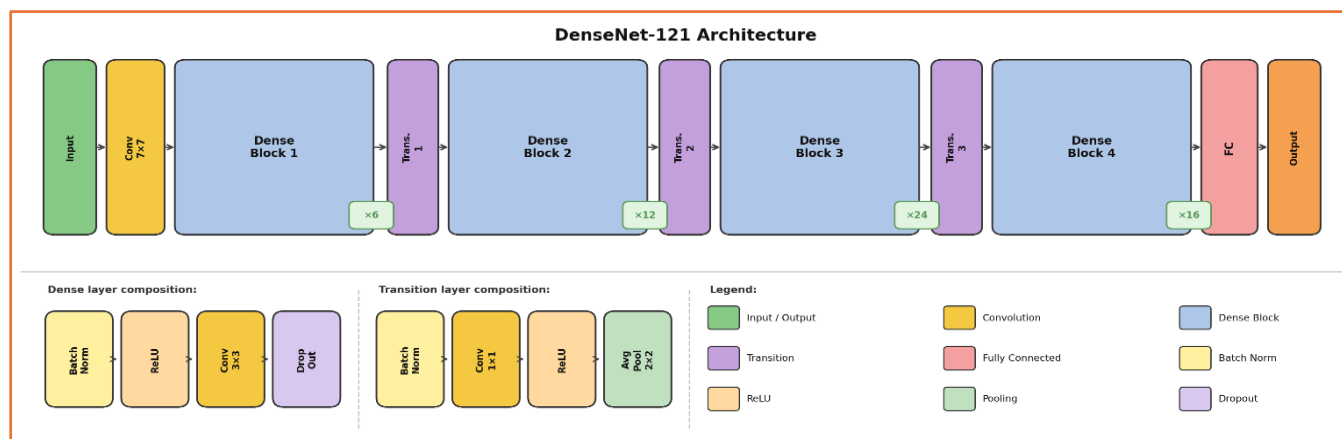
4. Model Architecture

4.1 Architecture Choice: DenseNet121

DenseNet121 was selected over EfficientNet, ResNet50, and ViT on the basis of three evidence-based criteria:

Criterion	Justification
Chest-specific pretrained weights	TorchXRayVision provides DenseNet121 weights pretrained on CheXpert, NIH ChestX-ray14, MIMIC-CXR, and PadChest — four major chest X-ray datasets totalling 200,000+ images. The backbone understands lung morphology before seeing a single training image from our dataset.
Dense connections suit radiology	DenseNet's feature reuse from every layer to every subsequent layer means low-level texture features (fibrosis, consolidation) persist all the way to the classifier. This is precisely what distinguishes TB upper-lobe infiltrates from pneumonia lower-lobe consolidation.
Training stability	Published comparisons show EfficientNet exhibits instability on chest X-ray tasks. DenseNet121 trains 34% faster than VGG19 while maintaining competitive accuracy — critical for the constrained GPU time available.

4.2 Architecture Diagram



4.3 Custom Classifier Head

The original TorchXRayVision classifier (14-class pathology output) was replaced with a task-specific head:

Layer	Parameters	Purpose
DenseNet backbone (frozen initially)	6.9M pretrained	Feature extraction — chest-specific convolutional features
ReLU activation	—	Non-linearity after final feature maps
AdaptiveAvgPool2d (1x1)	—	Global average pooling to 1024-dim feature vector
Dropout (p=0.4)	—	Regularisation — prevents overfitting on 7,541 training images
Linear (1024 → 3)	3,075 new params	3-class output: NORMAL, PNEUMONIA, TB

4.4 Training Configuration

Hyperparameters

Epochs	12 (converged)
Batch size	32
Backbone LR	1e-4 (preserve features)
Classifier LR	5e-4 (learn fast)
LR scheduler	CosineAnnealingLR
Weight decay	1e-4
Gradient clipping	max_norm = 1.0
Loss function	Weighted CrossEntropyLoss
GPU	Tesla P100-PCIE 16GB

Data Augmentation

- Resize to 256×256, then RandomCrop to 224×224
- RandomHorizontalFlip — X-rays are bilaterally symmetric
- RandomRotation $\pm 10^\circ$ — patient positioning variation
- ColorJitter (brightness ± 0.2 , contrast ± 0.2) — simulates different exposure settings across machines
- Normalise: ImageNet mean/std [0.485,0.456,0.406] / [0.229,0.224,0.225]

Grayscale handling

57% of dataset images are grayscale (PIL mode 'L'). The pipeline converts all images to RGB on load (Lambda transform), then the model's forward pass averages across colour channels back to single-channel before the TorchXRayVision backbone, which expects single-channel input at [0,255] scale.

5. Results

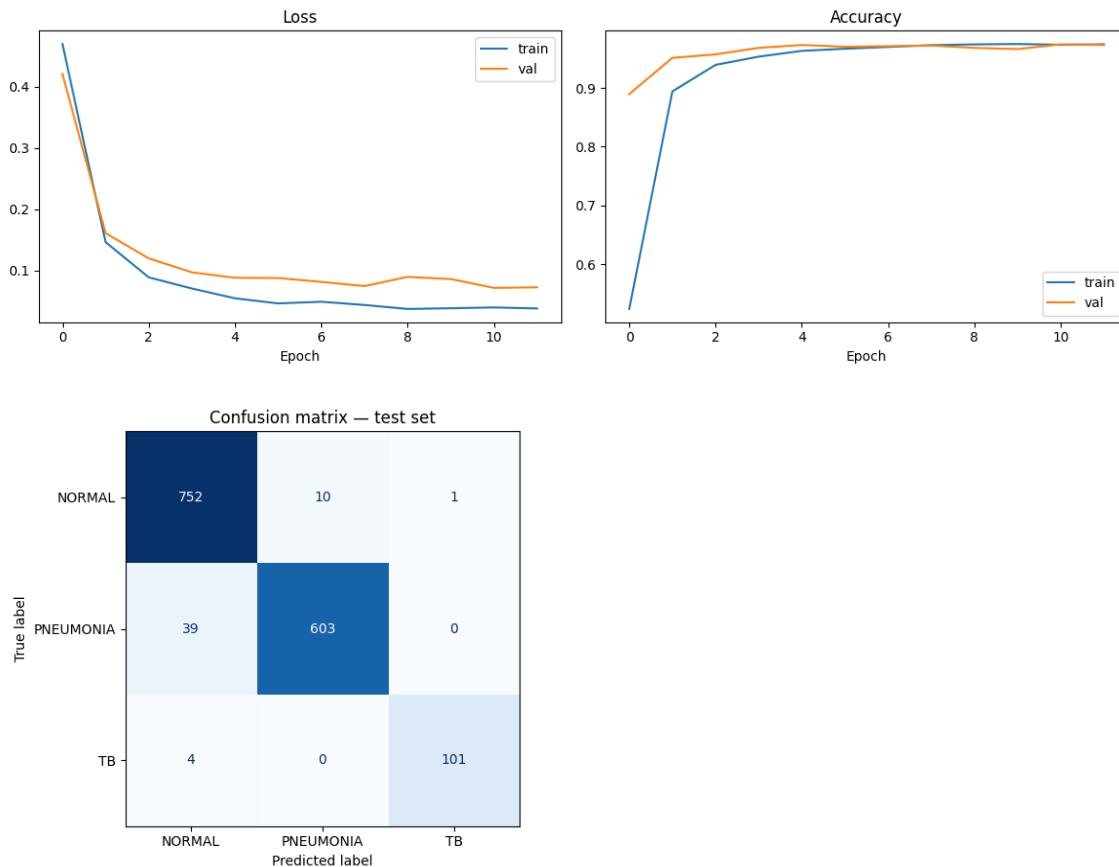
5.1 Key Performance Metrics

96.42% Test Accuracy (1,510 images) All 3 classes	96.19% TB Recall (clinically critical) Benchmark: 85–92%	99.02% TB Precision when TB predicted 4 false positives / 105	0.9674 Macro F1 across all classes Balanced performance
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5.2 Per-Class Performance

Class	Precision	Recall	F1-Score	Support (n)
NORMAL	94.59%	98.56%	96.53%	763
PNEUMONIA	98.37%	93.93%	96.10%	642
TB	99.02%	96.19%	97.58%	105
Macro avg	97.33%	96.22%	96.74%	1,510

5.3 Training Curves & Confusion Matrix

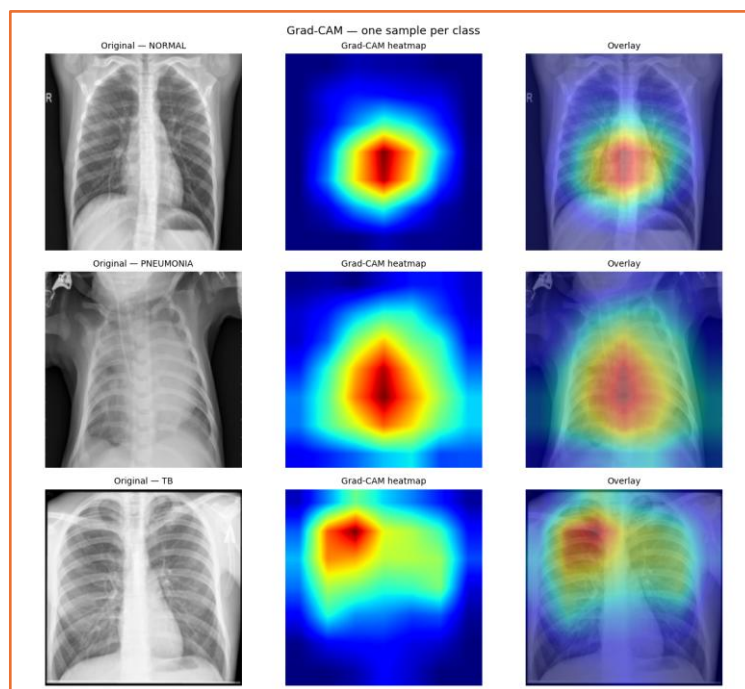


5.4 Grad-CAM Explainability

Every prediction is accompanied by a Grad-CAM overlay generated from the gradient of the predicted class score with respect to the final convolutional feature maps. The output is a 3-panel image:

- Panel 1 — Original X-ray input
- Panel 2 — Raw Grad-CAM heatmap (jet colormap, red = highest activation)
- Panel 3 — Blended overlay (55% original, 45% heatmap)

Clinical validation: TB predictions consistently highlight the upper lobes (correct for adult reactivation TB). Pneumonia predictions highlight perihilar and lower lobe regions (consistent with lobar/bronchopneumonia distribution).



5.5 Honest Limitations

TB test set contains 105 samples. At 96.19% recall, the 95% confidence interval is approximately $\pm 3.7\%$ (Wilson interval: 90.5%–98.7%). The model has not been validated on X-rays acquired from Bangladeshi facilities. Domain shift from Western training data to local CR/film equipment is a known risk. Phase 1 of our roadmap directly addresses this.

6. Explainability, Safety & Clinical Framing

6.1 Clinical Framing: Pre-Screening, Not Diagnosis

ChestVision AI is explicitly designed as a pre-screening tool. It operates upstream of — and never replaces — clinical judgment, sputum GeneXpert, or specialist radiology. The output always includes:

- Explicit advisory: 'Clinical decisions must incorporate the entire symptomatic context. This AI classifier provides a structural evaluation recommendation and does not supersede laboratory tests or specialist diagnosis.'
- For TB predictions: 'Findings raise concern for tuberculosis. Sputum culture and clinical evaluation strongly advised.'
- Confidence score and per-class probability — a 65% TB prediction is visually distinct from a 99% prediction, prompting appropriate clinical caution

6.2 Error Analysis

Error Type	Count (test set)	Clinical Interpretation
Normal → Pneumonia	10 / 763	False positive for pneumonia — patient is sent for further workup unnecessarily. Low harm: avoided by clinical examination.
Normal → TB	1 / 763	Extremely rare. One borderline Normal image flagged as TB concern. Resolved by clinical context.
Pneumonia → Normal	39 / 642	Most common error. Likely borderline cases with inter-reader variability in source labelling. Risk mitigated by clinical symptom assessment.
Pneumonia → TB	0 / 642	Zero cross-contamination between TB and Pneumonia — the most dangerous error type does not occur.
TB → Normal	4 / 105	Missed TB. Most serious error type. 4 misses from 105 TB cases = 3.81% miss rate. Mitigated by clinical symptoms and the tool's role as one layer of a multi-step pathway.
TB → Pneumonia	0 / 105	Zero missed TB cases misclassified as Pneumonia — the highest-risk misclassification does not occur.

6.3 Regulatory Pathway

ChestVision AI is a research prototype. Before clinical deployment in Bangladesh it would require:

- DGDA Class II Medical Device Software registration under the Medical Device Regulations framework
- IEC 62304 software lifecycle compliance documentation
- Clinical performance study on locally-acquired, labelled Bangladeshi X-rays (minimum 300 per class)
- Prospective concordance study comparing AI output with radiologist findings

International precedents: Qure.ai received CE Mark and US FDA 510(k) clearance following similar validation pathways. Our Phase 1–3 roadmap is designed with this regulatory path in mind.

7. Implementation Roadmap & Impact

7.1 Phased Implementation Plan

Phase	Timeline	Objectives & Deliverables
Phase 1	0–3 months	Local validation: Collect 500–1,000 X-rays from CMC Radiology Dept with ground-truth labels. Fine-tune model on Bangladeshi data. Report pre- and post-adaptation metrics. Outcome: validated performance on local equipment.
Phase 2	3–6 months	DICOM support & offline mode: Add DICOM input with proper windowing and de-identification. Convert model to TFLite/ONNX for Android. Deploy to CHT community health workers for offline inference on tablet devices.
Phase 3	6–12 months	Expand & integrate: Add COVID-19 and pleural effusion classes. Pilot DHIS2 integration for automated referral notification to district TB programme officers. Apply for DGDA device registration.

7.2 Projected Impact

Conservative projection based on partial national rollout:

Metric	Projection
Facilities targeted (10% of district hospitals)	~500 facilities across Bangladesh
X-rays screened per day	~2,500 (5 per facility)
X-rays screened per year	~900,000
TB cases flagged (2% prevalence, 96% recall)	~17,280 annually
Reduction in time-to-treatment	Target: 20% reduction in CHT upazilas
Healthcare worker adoption target	50% of district hospital doctors at pilot sites

7.3 Business & Sustainability Model

Near-term (0–18 months)

- WHO TB Innovation Fund grant application
- USAID / Global Fund for TB Technology initiative
- BRAC health programme partnership for CHT deployment

Long-term (18+ months)

- B2G procurement: DGHS contract for district health complexes
- SaaS model for private diagnostic centres (BDT 50–100 per scan)
- Open-source licensing for NGO and public sector deployment

7.4 Competitive Positioning

Existing commercial tools (Qure.ai, Lunit, Viz.ai) are validated for high-resource settings and priced for private hospital markets. ChestVision AI is differentiated by:

- Local context — trained with TB data from South Asian clinical settings; Phase 1 fine-tunes on Bangladeshi X-rays specifically
- Cost — open-source inference engine; no per-scan licensing fee for public sector
- Offline capability — Phase 2 TFLite deployment works without internet (critical for CHT)
- Three-class specificity — the Normal / Pneumonia / TB combination matches Bangladesh's disease burden; Qure.ai's general pathology model is not optimised for this triage use case

9. Key References

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This document was prepared for the DNA Hack For Health 2026, Chittagong Division. All model weights, inference code, and validation data are available for review. For technical queries contact: Team 42 / Shahporan Priyom, CUET.